Tail Leakge as a Confidence Signal

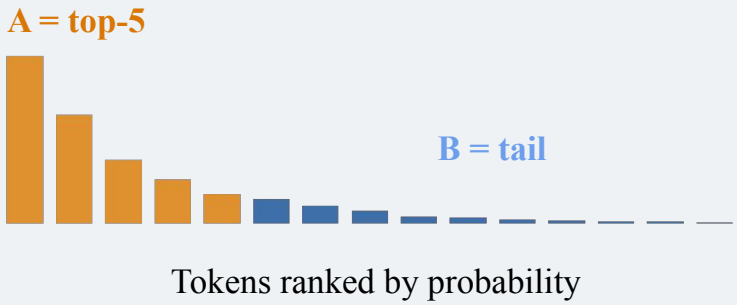
$$H(p) = -\sum_{v \in \mathcal{V}} p(v) \log p(v).$$

$$p(v) = \begin{cases} m_5 p_{\text{top5}}(v), & v \in \mathcal{V}_{\text{top5}}, \\ (1 - m_5) p_{\text{tail}}(v), & v \in \mathcal{V}_{\text{tail}}. \end{cases}$$

$$H(p) = -\sum_{v \in \mathcal{V}_{\text{top5}}} p(v) \log p(v) - \sum_{v \in \mathcal{V}_{\text{tail}}} p(v) \log p(v)$$

$$= -m_5 \log m_5 - (1 - m_5) \log(1 - m_5) + m_5 H(p_{\text{top5}}) + (1 - m_5) H(p_{\text{tail}})$$

Partition the candidate tokens into the top-five and tail sets based on their probabilities.



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Which Part of Entropy Detects Errors?

Error-detection AUROC, average across four datasets

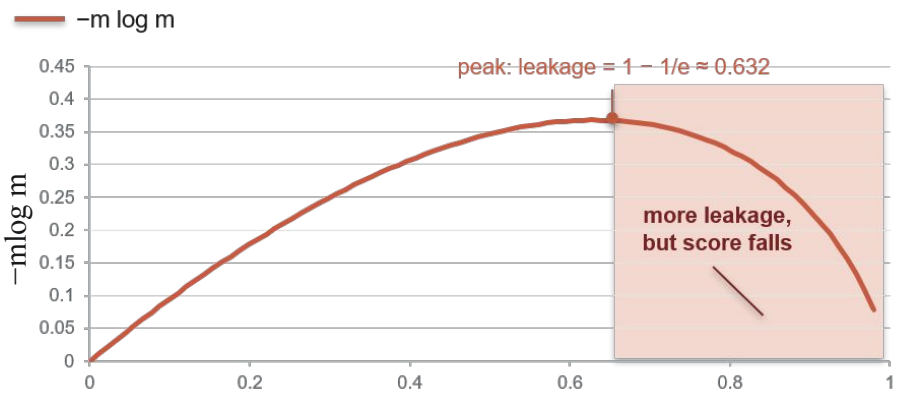
$-m_5 \log m_5 - (1 - m_5) \log(1 - m_5)$ **: Top-5 vs. tail mass split** — **0.742**

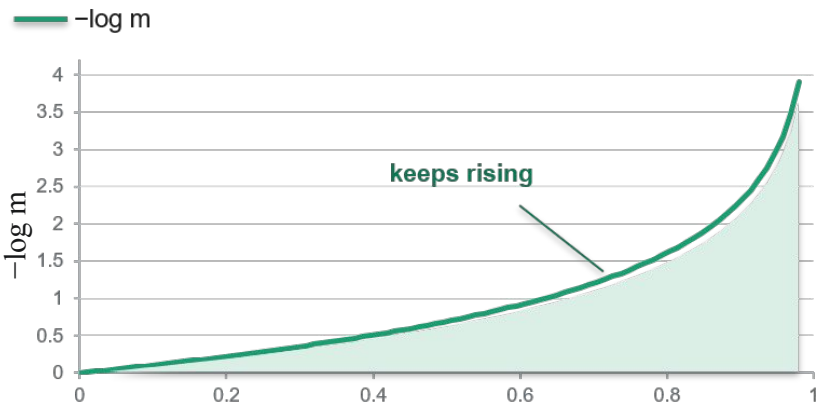
$H(p_{\text{top5}})$ **: Uncertainty within top-5** — 0.712

$H(p_{\text{tail}})$ **: Uncertainty within the tail** — 0.502

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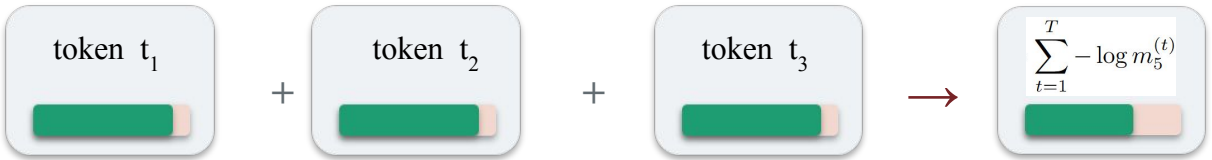
Tail Leakge as a Confidence Signal

$$H(p) = -\sum_{v \in \mathcal{V}} p(v) \log p(v).$$

$$= -m_5 \log m_5 - (1 - m_5) \log(1 - m_5) + m_5 H(p_{\text{top5}}) + (1 - m_5) H(p_{\text{tail}})$$

$$\longrightarrow \quad -\log m_5^{(t)}$$

$$\text{TailLeakage}(y) = \sum_{t=1}^{T} -\log m_5^{(t)}$$



Tail Leakge as a Confidence Signal

Higher Tail Leakge is associated with higher error rate.

